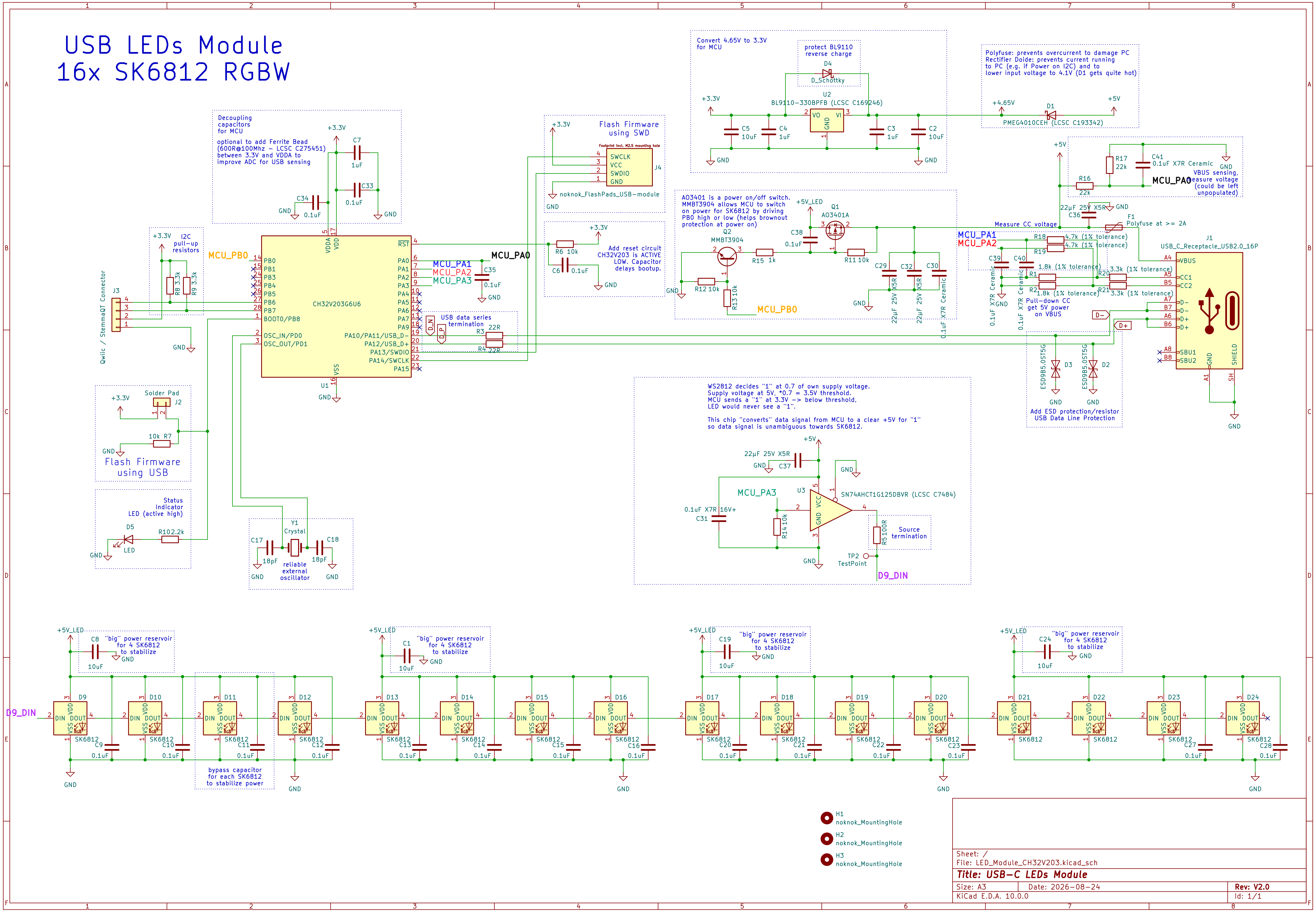
USB LEDs Module
16x SK6812 RGBW
Decoupling capacitors for MCU
optional to add Ferrite Bead (600R@100Mhz - LCSC C275451) between 3.3V and VDDA to improve ADC for USB sensing
Flash Firmware using SWD
noknok_FlashPads_USB-module
Add reset circuit CH32V203 is ACTIVE LOW. Capacitor delays bootup.
Convert 4.65V to 3.3V for MCU
protect BL9110 reverse charge
D_Schottky
BL9110-330BPFB (LCSC C169246)
Polyfuse: prevents overcurrent to damage PC
Rectifier Doide: prevents current running to PC (e.g. if Power on I2C) and to lower input voltage to 4.1V (D1 gets quite hot)
PMEG4010CEH (LCSC C193342)
VBUS sensing, measure voltage (could be left unpopulated)
AO3401 is a power on/off switch. MMBT3904 allows MCU to switch on power for SK6812 by driving PB0 high or low (helps brownout protection at power on)
Measure CC voltage
Polyfuse at >= 2A
USB_C_Receptacle_USB2.0_16P
Pull-down CC get 5V power on VBUS
Add ESD protection/resistor USB Data Line Protection
I2C pull-up resistors
Qwiic / StemmaQT Connector
CH32V203G6U6
USB data series termination
Solder Pad
Flash Firmware using USB
Status Indicator LED (active high)
reliable external oscillator
WS2812 decides "1" at 0.7 of own supply voltage. Supply voltage at 5V, *0.7 = 3.5V threshold. MCU sends a "1" at 3.3V -> below threshold, LED would never see a "1".
This chip "converts" data signal from MCU to a clear +5V for "1" so data signal is unambiguous towards SK6812.
SN74AHCT1G125DBVR (LCSC C7484)
Source termination
TP2 TestPoint
D9_DIN
"big" power reservoir for 4 SK6812 to stabilize
bypass capacitor for each SK6812 to stabilize power
noknok_MountingHole
Sheet: /
File: LED_Module_CH32V203.kicad_sch
Title: USB-C LEDs Module
Size: A3
Date: 2026-08-24
Rev: V2.0
KiCad E.D.A. 10.0.0
Id: 1/1